

SHAH INVESTOR'S HOME LIMITED

Synapse

Business Requirements Specification

Document	Business Requirements Specification (BRS)
System	Synapse — centralised access and reporting platform
Organisation	Shah Investor's Home Limited (SIHL), SEBI-registered stockbroker
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1. Purpose and Scope

1.1 Purpose

Synapse is an internal web platform that gives SIHL staff a **single sign-on entry point to several business applications**, together with a reporting and data-entry engine that business users can extend *without software development*.

Today, adding a new report to an internal system typically requires a development cycle: specification, code change, testing, release. Synapse removes that dependency for a large class of requirements. An authorised administrator composes a report from reusable filter components and a query definition, and it becomes available to the intended users immediately.

1.2 In scope

- Centralised authentication with password policy, login auditing and optional IP restriction.
- A multi-application portal: one login, several products, each with its own menu structure.
- Role-based authorisation down to individual menu options.
- An administrator-driven reporting engine, including filters, formatting, totals, export and drill-down.
- The same engine reused as a data-entry form for adding, editing and deleting records.
- Internal notifications, targeted at a hierarchy level or an individual, with optional scheduling.
- Administration screens for every master entity the platform depends on.
- Usage analytics driving a "frequently used options" shortcut list.

1.3 Out of scope

- Outbound email or SMS delivery. Requests are captured for follow-up; no gateway is integrated.
- Client-facing (external) access. All users in this phase are internal.
- Trading, order management, settlement or accounting functionality. Synapse reports on and administers data; it does not execute business transactions.
- Migration of existing reports from legacy systems.

2. Business Context

SIHL operates across multiple business lines and a distribution network of branches, franchisees, sub-brokers and remisiers. Staff in different parts of the organisation need access to different systems and different subsets of data, governed by both the **company** they belong to and their position in the organisational **hierarchy**.

Two recurring problems motivate this platform:

- **Fragmented access.** Separate logins per system create credential sprawl, inconsistent access revocation when staff move or leave, and no single view of who can reach what.

- **Reporting bottleneck.** Every new report or data-capture screen competes for development capacity, so operational teams wait weeks for information they need routinely.

Synapse addresses both: one identity and authorisation model shared across applications, and a configuration-driven engine that shifts routine report creation from developers to administrators.

3. Stakeholders and User Roles

Users are classified by **hierarchy**, which is a distinct concept from **role**. Hierarchy describes *who the person is* within the organisation; role describes *what they may do*. Authorisation is evaluated against the combination of company, hierarchy and role.

Hierarchy	Code	Typical interest in the platform
Management	0000	Consolidated performance reporting and oversight
CSO	0100	Customer servicing, client queries, operational reports
Employee	0200	Day-to-day operational reports and data entry
Branch	0300	Branch-level business performance
Franchisee	0400	Own-network business and payout information
Sub-broker	0500	Own-network business and payout information
Remisier	0600	Own-network business and payout information
Family	0700	Grouped client relationships
Client	0800	Reserved for a later client-facing phase
Admin	9999	Platform administration and configuration

Note. A user holds one hierarchy but may hold several roles. Menus are granted to roles, and a user receives the union of the menus granted to their active roles, restricted to their own company and hierarchy.

4. Business Objectives

Ref	Objective	Success indicator
OBJ-1	Provide one login across all internal applications	A user reaches every application they are entitled to without re-authenticating
OBJ-2	Remove development dependency for routine reports	New reports are published by an administrator without a code release

OBJ-3	Make access rights auditable and revocable in one place	Removing a role immediately withdraws the corresponding menus and applications
OBJ-4	Ensure users only see data they are entitled to	Reports are scoped by company and hierarchy; user-entered records are private to their author
OBJ-5	Reduce the effort of reaching frequently needed screens	Favourites and usage-based shortcuts on the landing page
OBJ-6	Communicate operational messages reliably to defined audiences	Notifications reach a hierarchy or an individual, with read tracking

5. Functional Requirements

5.1 Authentication and access control

Ref	Requirement	Priority
FR-1.1	Users authenticate with a username and password.	Must
FR-1.2	Passwords must meet a defined complexity policy: minimum eight characters, at least one uppercase letter, one digit and one special character, and must not equal the username.	Must
FR-1.3	A new password must not match any of the user's recent previous passwords.	Must
FR-1.4	Every login attempt, successful or failed, is recorded with timestamp, outcome, failure reason, IP address and user agent.	Must
FR-1.5	A user may optionally be restricted to one or more IP address patterns. A user with no active restriction may log in from any network.	Must
FR-1.6	Sessions expire automatically after a defined period of validity.	Must
FR-1.7	Users may request assistance for a forgotten user ID or a password reset. These requests are captured for staff follow-up.	Should
FR-1.8	Administrators may reset a user's password directly.	Must

5.2 Multi-application portal

Ref	Requirement	Priority
FR-2.1	The platform hosts multiple applications, each belonging to a company, each with its own name, icon or logo and display order.	Must
FR-2.2	An application switcher in the top-right corner lists the applications the signed-in user may open, showing each application's logo and name.	Must
FR-2.3	Selecting an application replaces the menu structure with that application's menus.	Must
FR-2.4	The name of the currently open application is displayed in the top bar, visually separated from the product name and the search field.	Must
FR-2.5	The system remembers the last application a user opened and reopens it at their next login.	Must
FR-2.6	When the user has enabled the "new window" preference, selecting an application opens it in a new browser tab.	Should
FR-2.7	Menu search, favourites and usage shortcuts reflect only the currently open application.	Must

5.3 Navigation and personalisation

Ref	Requirement	Priority
FR-3.1	Menus support three levels: main menu, sub-menu and menu option.	Must
FR-3.2	A menu option may open an internal screen, a configured report, or an external web address.	Must
FR-3.3	The menu panel can be collapsed to an icon rail to maximise working area; collapsed items reveal their children on selection.	Must
FR-3.4	Opening a main menu closes any other open main menu, so only one section is expanded at a time.	Should
FR-3.5	The currently selected menu option is visually indicated.	Should
FR-3.6	A search field locates menu options by name or by their position in the menu structure, showing the full path for each match.	Must
FR-3.7	Users may mark any menu option as a favourite and reach it from the dashboard.	Should
FR-3.8	Users may choose a visual theme, a vertical or horizontal menu layout, and an interface language.	Should
FR-3.9	The interface is available in English and nine Indian regional languages.	Should

5.4 Dynamic reporting

Ref	Requirement	Priority
FR-4.1	Administrators define reusable filter components — text, number, date, date range, dropdown, multi-select, checkbox and radio — once, and reuse them across many reports.	Must
FR-4.2	Dropdown options may be a fixed list or drawn from a query, and one dropdown may depend on another so that its options narrow accordingly.	Must
FR-4.3	Administrators arrange filter components into a named filter, controlling row and position, mandatory status, label and default value.	Must
FR-4.4	A report binds a filter to a query definition and to display settings.	Must
FR-4.5	Mandatory filters must be supplied before a report will run.	Must
FR-4.6	Date ranges are validated: the start date may not be in the future and the end date must follow the start date.	Must
FR-4.7	Each column has a declared data type controlling its formatting and alignment: dates as dd-Mmm-yyyy , numbers right-aligned with configurable decimal places, text left-aligned.	Must
FR-4.8	Results can be sorted by any column, with dates and numbers sorted by value rather than by their displayed text.	Must

FR-4.9	Selected columns may display a total in the footer.	Should
FR-4.10	Rows may be highlighted automatically according to configured conditions.	Should
FR-4.11	Leading columns may be frozen so they remain visible while scrolling horizontally.	Should
FR-4.12	If a result exceeds the configured maximum row count, the report is not rendered on screen; the user is offered a download instead.	Must
FR-4.13	Results can be exported to CSV, named consistently as {username}_{timestamp}.csv .	Must
FR-4.14	The parameter panel can be hidden to give the results the full working area, and its state is indicated on the breadcrumb line.	Should
FR-4.15	A column may be configured to drill into another report, either as a full page with a back link or as a popup window.	Should
FR-4.16	Returning from a drill-down restores the originating report exactly as it was left, including its filter values and results.	Must

5.5 Dynamic data entry

Ref	Requirement	Priority
FR-5.1	The same filter definition may be used as a data-entry form for adding, editing and deleting records in a target table.	Must
FR-5.2	Each form field is mapped to a column in the target table.	Must
FR-5.3	Records created through a form are visible and editable only to the user who created them.	Must
FR-5.4	Every record records who created it, who last updated it and when.	Must
FR-5.5	A user's own saved records are listed beneath the form, with edit and delete actions; deletion requires confirmation.	Must

5.6 Notifications and messaging

Ref	Requirement	Priority
FR-6.1	Administrators compose messages addressed either to an entire hierarchy or to one named individual.	Must
FR-6.2	A message may be delivered as a popup on the recipient's next sign-in, as an entry in the notification bell list, or both.	Must
FR-6.3	A message may be scheduled; recipients do not see it before the scheduled date and time.	Must
FR-6.4	The notification bell shows a count of unread messages and lists messages with their timestamps.	Must

FR-6.5	Read status is tracked per recipient.	Must
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5.7 Administration

Ref	Requirement	Priority
FR-7.1	Administrators maintain the following masters: Company, App, Hierarchy, Client Category, User IP Mapping, Menu, Role, User, Filter Components, Filters, Reports and Notifications.	Must
FR-7.2	Every administration screen supports create, edit and delete, with validation messages shown against the offending field.	Must
FR-7.3	Every administration screen provides a free-text search and, where applicable, a filter by company.	Must
FR-7.4	Administration lists can be exported to CSV, reflecting the current search and filter selection.	Should
FR-7.5	Menus are assigned to roles, and roles are assigned to users, each through a checklist that shows enough detail to identify the item.	Must
FR-7.6	Fields that do not apply to the current selection are hidden rather than shown as inapplicable.	Should
FR-7.7	Reference fields are displayed by name rather than by code wherever a name exists.	Should

5.8 Usage analytics

Ref	Requirement	Priority
FR-8.1	Each menu selection is recorded with its timestamp, company, application, main menu, sub-menu and menu option.	Must
FR-8.2	Usage records are retained for seven days and removed automatically thereafter.	Must
FR-8.3	The dashboard shows the five most frequently used options, beneath Favourites and presented in the same style.	Must
FR-8.4	Recording usage must not delay or interfere with navigation.	Must

6. Business Rules

These rules govern behaviour that is not self-evident from the requirements above and must be preserved in any future change.

Ref	Rule	Rationale
BR-1	A user's access to an application is derived , not assigned. A user may open an application if any menu they are entitled to belongs to that application.	Removes a second permission list that could contradict menu permissions. Granting a menu is sufficient and unambiguous.
BR-2	A menu that is not assigned to any application is global and appears in every application. A global menu does not itself grant access to any application.	Administration must remain reachable regardless of the application in use; otherwise an administrator could switch application and lose the ability to switch back.
BR-3	Menu permissions are evaluated as the union of the user's active roles, restricted to the user's own company and hierarchy.	The same role code may exist for several companies or hierarchies with different meaning.
BR-4	Records created through a data-entry form are visible only to their creator. The restriction is enforced by the system, not by report configuration.	A report author cannot accidentally expose another user's records by omitting a condition.
BR-5	A scheduled notification is simply not visible before its scheduled time; no separate send action exists.	Avoids a background dispatch process and the failure modes that come with it.
BR-6	Usage records older than seven days are deleted. Reporting on usage never considers records outside that window.	Satisfies the retention limit even if housekeeping is delayed.
BR-7	Deactivating an application, or withdrawing the role that granted it, silently returns the user to their first available application at the next sign-in.	A withdrawn entitlement must never leave a user unable to sign in.
BR-8	Where a summary report and its detail report are intended to share filter values, both must use the same filter component. Values carry across automatically; there is no separate mapping to maintain.	Keeps drill-down configuration to a single decision and prevents mismatched mappings.
BR-9	Downloaded files are always named {username}_{timestamp}.	Makes the origin of a shared file identifiable.

7. Non-Functional Requirements

Ref	Category	Requirement
NFR-1	Security	Passwords are stored only as salted hashes and are never recoverable or displayed.
NFR-2	Security	Values entered by users are never combined textually into database queries.
NFR-3	Security	Only administrators may create or change report and filter definitions.
NFR-4	Security	Session identifiers are held in a cookie inaccessible to page scripts.
NFR-5	Auditability	Login attempts and record ownership are retained for review. Usage analytics are retained for seven days only.
NFR-6	Performance	Menu and application permissions are resolved once per page request.
NFR-7	Performance	Recording menu usage must not add perceptible delay to navigation.
NFR-8	Performance	Reports must not attempt to render result sets beyond their configured maximum; an export path is offered instead.
NFR-9	Usability	Navigation must keep the top bar and menu continuously visible and usable while a screen loads.
NFR-10	Usability	Dates are presented consistently as <code>dd-Mmm-yyyy</code> throughout, avoiding ambiguity between date conventions.
NFR-11	Maintainability	Adding a routine report must not require a software release.
NFR-12	Compatibility	The platform targets current versions of mainstream desktop browsers.

8. Assumptions, Dependencies and Constraints

8.1 Assumptions

- All users in this phase are internal staff with organisation-issued credentials.
- Reporting queries are authored by administrators who understand the underlying data and are trusted to do so.
- Users have reliable network access to the internal environment.

8.2 Dependencies

- A PostgreSQL database holding both platform configuration and the business data being reported on.
- Company and hierarchy reference data must be populated before users can be created.
- Any table used for data entry must carry the ownership and audit columns required by BR-4.

8.3 Constraints

- No outbound email or SMS gateway is available; assistance requests are captured only.
- Menu names, report titles and other administrator-authored text appear as entered and are not translated by the language setting.
- Usage analytics cannot answer questions spanning more than seven days.

9. Future Phases

Ref	Candidate capability	Note
FUT-1	Asynchronous export queue with Excel and PDF output, and a download centre	Design allows for it; current export is synchronous CSV
FUT-2	Email and SMS delivery of notifications and scheduled reports	Requires a gateway decision
FUT-3	Client-facing access for the Client hierarchy	Hierarchy already defined
FUT-4	Dashboard analytics from live login and usage data	Dashboard currently shows indicative figures
FUT-5	Organisation-wide usage reporting for administrators	Data is already captured; only the presentation is user-specific today
FUT-6	Longer-term usage retention for trend analysis	Requires a retention policy decision

10. Glossary

Term	Meaning
Application (App)	One product within a company, selectable from the application switcher, with its own menu structure.
Company	The legal entity a user, menu, role or application belongs to.
Hierarchy	The user's position in the organisation, such as Branch or Franchisee. Distinct from role.
Role	A named set of menu permissions, scoped to a company and hierarchy.
Filter component	A reusable input definition, such as "Company" or "Sale Date Range", used across many reports.
Filter	A named arrangement of filter components into rows and positions.
Report	A filter bound to a query definition and display settings.
Report mode	A report that runs a query and displays results.
Form mode	The same filter used to add, edit and delete records in a target table.

Drill-down	Opening a related report from a value in a result row, carrying the current filter values across.
Global menu	A menu not assigned to any application, therefore visible in all of them.

SIHL Synapse — Business Requirements Specification, version 1.0. Companion documents: Technical Architecture; User Guidelines.